

CURRICULUM VITAE

	Rajeev Ranjan Institute Assistantship Research Scholar (IARS) Department of Chemical Engineering Indian Institute of Technology Roorkee, Roorkee-247667 Uttarakhand, India Email: rajeev_r@ch.iitr.ac.in , rajeev.raja.737@gmail.com Phone No: +91-9910947764
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Academic background

July 2021 – Thesis submitted Ph.D. (Chemical Engineering) Department of Chemical Engineering Indian Institute of Technology Roorkee, Roorkee-247667, Uttarakhand, India CPI: 8.66/10	August 2019 - June 2021 M.Tech (Chemical Engineering) Department of Chemical Engineering University College of Science & Technology, University of Calcutta, West Bengal, India CPI: 8.86/10
August 2015 - June 2019 B.Tech (Chemical Engineering) Department of Chemical Engineering University College of Science & Technology, University of Calcutta, West Bengal, India CPI: 8.32/10	2015: Intermediate (Class XII) B.S.S.N. Inter College, Deo, Aurangabad, India Percentage: 72.8 2012: Matriculate (Class X) Jawahar Navodaya Vidyalaya CGPA: 10/10

Research profile (Current research area)

- Design and synthesize of a nano-structured catalyst for CO₂ hydrogenation to selective formic acid.
- Process optimization using response surface methodology (RSM).
- Kinetic modelling for heterogeneous reactions.
- Life cycle assessment (LCA) of the CO₂ hydrogenation process to evaluate environmental impact.

Ph.D. thesis title: CO₂ hydrogenation to formic acid

- Design and synthesize of a nano-structured Ni-based catalyst supported on MgO, γ -Al₂O₃, and MOF for selective CO₂ hydrogenation to formic acid.
- Characterization of the synthesized catalyst by X-ray diffraction study, specific surface area measurement, temperature-programmed reduction, temperature-programmed oxidation, temperature programmed desorption, H₂-pulse chemisorption study, scan electron microscopy, transmission electron microscopy, thermal gravimetric analysis, etc.
- Process optimization using response surface methodology (RSM) for maximizing FA yield.
- Kinetic modelling of heterogeneous reactions to predict reaction dynamics and mechanisms.

Technical skills:

Familiar with various analytical instruments like gas chromatography, [HPLC](#), micromeritics pulse chemisorbs 2705 equipment ([BET](#), [TPR](#), [TPD](#), [TPO](#), [chemisorption study](#)), [BET equipment](#) (catalyst surface area, porosity, pore-volume, and pore size measurement), [XRD](#), [SEM](#), [TEM](#), [TGA](#), etc.

Ph.D. thesis supervisor: Dr. Prakash Biswas, Professor, Department of Chemical Engineering, Indian Institute of Technology Roorkee, 247667, Uttarakhand, India. (CO₂ utilization to formic acid)

M.Tech thesis supervisor: Dr. Subhasis Roy, Assistant Professor, Department of Chemical Engineering, University of Calcutta. (Fabrication of a Perovskite solar cell)

B.Tech thesis supervisor: Dr. Kaushik Ganguly, Associate Professor, Department of Chemical Engineering, University of Calcutta. (An investigation on Turbulence Induced Interfacial Precipitation)

PUBLICATIONS

Journal Publications:

1. Rajeev Ranjan, Prakash Biswas*, “CO₂ reduction to formic acid via in-situ hydride generation from NaBH₄: Process parameter modelling through ANN-based approach”, International Journal of Hydrogen Energy, 242 (2026) 155663, **Elsevier**, Impact factor 8.3. **Rank Q1**
[DOI: 10.1016/j.ijhydene.2026.155663](https://doi.org/10.1016/j.ijhydene.2026.155663)
2. Rajeev Ranjan, Prakash Biswas*, and K. K. Pant, “Study on the interaction effect of reaction parameters for the hydrogenation of CO₂ to formic acid (FA) over a Ni-alumina-MOF (Ni-M_A-520) catalyst in an aqueous medium”, Catalysis Today, 460 (2025) 115489, **Elsevier**, Impact factor 5.3. **Rank Q1**
[DOI: 10.1016/j.cattod.2025.115489](https://doi.org/10.1016/j.cattod.2025.115489)
3. Rajeev Ranjan, Prakash Biswas*, and K. K. Pant, “Selective hydrogenation of CO₂ to formic acid with higher yield in an aqueous medium with a nano-nickel-metal catalyst: reaction parameter optimization by response surface methodology (RSM)”, Reaction Chemistry & Engineering, 9 (2024) 2635–2646, **RSC**, Impact factor 3.8. **Rank Q1**
[DOI: 10.1039/D4RE00194J](https://doi.org/10.1039/D4RE00194J)
4. Rajeev Ranjan, Prakash Biswas *, “Preparation and characterization of Ni/Al₂O₃ catalyst for catalytic reduction of CO₂ to formic acid in the presence of hydrazine hydrate as a hydrogen source” International Journal of Chemical Reactor Engineering 7 (2024) 835-841, **De Gruyter**, Impact factor 1.4. **Rank Q3**
[DOI: 10.1515/ijcre-2024-0038](https://doi.org/10.1515/ijcre-2024-0038)

Conference Proceedings:

1. Rajeev Ranjan & Subhasis Roy* “Supercapacitor electrodes: A mini review”
IAMWCASET – 2020 (ISBN Number: 978-81-945642-4-9)

Conference (National/International):

1. Rajeev Ranjan, Prakash Biswas*, “Sustainable pathway for CO₂ conversion into value-added chemicals in a circular economy”, **India International Science Festival**, 6-9th December 2025, Panchkula, Haryana, India.
2. Rajeev Ranjan, Prakash Biswas* “Role of catalyst supports in enhancing selective hydrogenation of CO₂ to formic acid over Ni-based catalysts”, **HES 2025 conference**, 10-12th October 2025, IIT BHU, India.
3. Rajeev Ranjan and Prakash Biswas* “Heterogeneous catalysis in aqueous media: Optimizing CO₂ hydrogenation to formic acid with metal-based nano catalysts” **The World Conference on Carbon 2025**, Saint-Malo, France, June 29 -July 4, 2025.
4. Rajeev Ranjan and Prakash Biswas* “Efficient and sustainable CO₂ hydrogenation to formic acid using a metal-based catalyst” **The 9th Green and Sustainable Chemistry Conference**, Pune, India, March 4-6, 2025.
5. Rajeev Ranjan and Prakash Biswas* “Sustainable conversion of CO₂ into valuable chemicals for a circular economy” **International Symposium on Transforming Waste Management: Circular Economy & Energy Sustainability**, SERB, IIT Roorkee, India, November 28-29, 2024. ([Certificate of Excellence Award](#))
6. Rajeev Ranjan and Prakash Biswas* “Conversion of CO₂ to formic acid: reaction parameters optimization by response surface methodology (RSM)” **International Conference on Advanced Functional Materials 2024 (ICAFM 2024)**, Kathmandu, Nepal, October 25-27, 2024.
7. Rajeev Ranjan and Prakash Biswas* “Optimizing reaction parameters with Ni/AC catalyst for catalytic reduction of CO₂ to formic acid and other valuable products” **ICECEES-2024**, Department of Chemical Engineering, IIT Roorkee, India, February 15-17, 2024.
8. Rajeev Ranjan and Prakash Biswas* “Hydrogenation of CO₂ to formic acid: An optimization of reaction parameters by RSM” **IChE-CHECON 2023**, Heritage Institute of Technology, Kolkata, India, December 27-30, 2023.
9. Rajeev Ranjan and Prakash Biswas* “Catalytic reduction of CO₂ to formic acid in the presence of hydrazine hydrate as a hydrogen source” **CUChEAA Symposium 2023**, Department of Chemical Engineering, University of Calcutta, Kolkata, India, December 22, 2023.

10. Rajeev Ranjan and Prakash Biswas* “Catalytic hydrogenation of CO₂ into formic acid and other valuable products” **International Conference on Chemical Engineering: Enabling Transition Towards Sustainable Future**, Department of Chemical Engineering, IIT Roorkee, India, September 8-10, 2022.
11. Rajeev Ranjan and Prakash Biswas* Participated in “**The World Petroleum Technology Congress-2022**”, The World Oil Spill Conference-2022, Vigyan Bhawan, New Delhi, India, December 7-8, 2022.

INDUSTRIAL EXPERIENCE

Oil and Natural Gas Corporation of India, Navi Mumbai, Maharashtra

Designation: Vocational Trainee (Research & Development) [June 15 –July 15]

- The main objective of the training program was to study the complete operation of an elevated flare system in the Oil and Gas industry and to prepare a comprehensive spreadsheet detailing its various components (Flare height, Diameter, Flame Length, etc.).

Responsibility:

- Manual reading of API 537 and 541 for understanding the working of an elevated flare.
- Preparing a complete working spreadsheet of the flare system, which includes calculations like Flare height, Flare diameter, flame length, etc.
- Simulating an operational prototype of any plant in ASPEN software.

ACADEMIC EXPERIENCE

- Short-term Course by IIT KGP “Fluid Mechanics for Biomimetic Lab on a Chip”
- Participated in ROBOTRYST-2016, organized by IIT Delhi.
- Entrepreneurship Development Workshop organized by the Department of Chemical Engineering, University of Calcutta.

Software skills: AutoCAD 2018, MS Office, Aspen HYSIS

CONTACT DETAILS

Present Address:

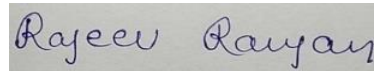
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Aurangabad, Bihar, 824202
India

SELF DECLARATION

I declare that the above information and facts stated are true, correct, and complete to the best of my knowledge and belief.



RAJEEV RANJAN

References:

1. Prof. Prakash Biswas

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2. Prof. B. Prasad

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